

Net Zero Pathways for Korea’s Petrochemical Industry: A Facility-Level Marginal Abatement Cost Analysis

PLANiT Institute
Seoul, Republic of Korea

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Abstract

This study presents a comprehensive facility-level marginal abatement cost curve (MACC) analysis for decarbonizing Korea’s petrochemical industry from 2025 to 2050. Using a database of 237 baseline facilities representing 100,066 kt/year production capacity, we evaluate six decarbonization scenarios combining three production pathways (Shaheen growth, 25% NCC restructuring, 40% NCC restructuring) with two naphtha cracking center (NCC) technology options (green hydrogen vs. electrification). Our analysis demonstrates that net-zero emissions by 2050 is technically and economically achievable through a portfolio approach combining electrification, green hydrogen, and renewable energy procurement. Baseline emissions of 47.0 MtCO₂/year at 70% operating rate can be fully eliminated with cumulative investment ranging from \$21.8B to \$41.7B, requiring up to 282 TWh/year of renewable electricity by 2050. Key findings include the critical synergy between heat pump deployment and RE-PPA procurement for non-NCC facility decarbonization, and the cost-effectiveness advantage of NCC electrification over hydrogen pathways.

Keywords: Petrochemical industry, Decarbonization, MACC, Net Zero, Electrification, Green Hydrogen, Korea

1 Introduction

Korea’s petrochemical industry ranks 4th globally in ethylene production capacity, making it a critical sector for the nation’s carbon neutrality goals. The industry produces approximately 47 MtCO₂/year at current operating rates (70%), representing a significant share of Korea’s industrial emissions. Under Korea’s 2050 Carbon Neutrality Strategy,

the sector faces substantial decarbonization pressure while maintaining economic competitiveness in a challenging global market characterized by overcapacity, particularly from China.

This study develops a facility-level MACC model to evaluate decarbonization pathways, addressing three key research questions:

1. What technology portfolio can achieve net-zero emissions by 2050?
2. What are the investment requirements and energy infrastructure needs?
3. How do different production scenarios affect abatement costs?

2 Methodology

2.1 Data Architecture

The model is driven by a completely externalized data architecture, ensuring transparency and reproducibility. The core logic relies on three primary data configurations:

2.1.1 Facility Database

The analysis covers 239 facilities (243 including the Shaheen project) across Korea’s four major petrochemical complexes (Yeosu, Ulsan, Daesan, and others). The database is sourced from the Korea Petrochemical Industry Association (KPIA) and company reports, detailing:

- **Production Capacity:** 100,066 kt/year (baseline).
- **Process Type:** Naphtha Cracker, BTX, Polymer, etc.
- **Energy Intensity:** Specific energy consumption (SEC) derived from KEEI and LBNL benchmarks.
- **Location:** Mapped to regional grids for precise infrastructure planning.

2.1.2 Scenario Configuration

Scenarios are defined via `scenario_definitions.csv`, enabling modular combination of production pathways (Shaheen Growth, Restructuring) and technology choices (Hydrogen vs Electrification).

2.1.3 Emission Targets

The Net Zero pathway is governed by `emission_targets.csv`, which sets the national reduction mandate: 15% (2030), 24.5% (2035), 50% (2040), 75% (2045), and 100% (2050).

2.2 Intertemporal Optimization Model

Unlike static Marginal Abatement Cost (MAC) models that optimize for a single snapshot year, this study employs a **Dynamic Recursive Optimization** algorithm to minimize the Net Present Value (NPV) of system costs over the 2025–2050 transition period.

2.2.1 Levelized Cost of Abatement (LCOA)

The core decision metric is the Levelized Cost of Abatement (LCOA), calculated dynamically for every candidate facility in every simulation year t :

$$LCOA_{i,t} = \frac{\sum_{n=t}^{2050} \frac{Cost_{i,n}}{(1+r)^{n-t}}}{\sum_{n=t}^{2050} Abatement_{i,n}} \quad (1)$$

Where:

- $Cost_{i,n} = CAPEX_{ann} + OPEX + Cost_{energy} - Cost_{fuel_saved}$
- $Abatement_{i,n}$ is the annual CO₂ reduction (tons).
- r is the social discount rate (8%).

This look-ahead metric ensures that technology deployment decisions made today account for future cost declines (learning curves) and rising carbon/energy prices.

2.2.2 Dynamic Deployment Algorithm

The optimization proceeds as follows:

1. **Check Gap:** In each year t , compare projected baseline emissions against the Net Zero target.
2. **Rank Candidates:** If a gap exists, calculate $LCOA_{i,t}$ for all undeployed facilities.
3. **Deploy:** Select facilities with the lowest LCOA until the emission gap is closed.
4. **Update State:** Deployed facilities remain decarbonized for their lifetime; the grid emission factor updates dynamically.

2.3 LCOH Methodology

Green hydrogen price is dynamically calculated using the Levelized Cost of Hydrogen formula:

$$LCOH = \frac{CAPEX \times CRF + OPEX + Stack_{repl}}{H_2 \text{ Production}} + Elec_{cost} \quad (2)$$

Key parameters include electrolyzer CAPEX declining from \$1,000/kW (2025) to \$500/kW (2050), efficiency improving from 70% to 75%, and 90% capacity factor. This methodology links hydrogen prices to RE prices, reflecting the true cost dynamics of green hydrogen production.

2.4 Emission Factors

Table 1: Combustion Emission Factors

Fuel	tCO ₂ /GJ	Source
Naphtha	0.0542	IPCC 2019
LNG	0.0561	IPCC 2019
Fuel Gas	0.050	API Compendium 2021
Byproduct Gas	0.048	API Compendium 2021
LPG	0.0631	IPCC 2019
Fuel Oil	0.0773	IPCC 2019

2.5 Price Trajectories

Table 2: Energy Price Trajectories

Parameter	2025	2030	2035	2040	2045	2050
RE Price (\$/MWh)	65	56	48	41	35	30
H ₂ Price (\$/kg)*	4.58	3.91	3.33	2.82	2.39	2.01
Grid EF (tCO ₂ /MWh)	0.436	0.349	0.280	0.140	0.070	0.000

*H₂ price calculated via LCOH formula. Source: IRENA 2024, IEA WEO 2024, PLANiT Institute

2.6 Key Assumptions

- **Operating Rate:** A constant 70% operating rate is assumed 2025–2050, reflecting structural overcapacity and market competition.
- **Learning Curves:** Technology costs (CAPEX) for H₂ furnaces, electric crackers, and electrolyzers are assumed to decline by 50% by 2050 relative to 2025, driven by global deployment and manufacturing scale.
- **Financials:** Real 2024 USD terms; 8% social discount rate; 25-year facility lifetime.

3 Scenario Design

3.1 Six Combined Scenarios

The study evaluates six scenarios combining three production pathways with two NCC technology options:

Table 3: Scenario Definitions

#	Scenario ID	Production Pathway	NCC Technology
1	shaheen_ncc_h2	Shaheen (Growth)	NCC-H ₂
2	shaheen_ncc_elec	Shaheen (Growth)	NCC-Electricity
3	restructure_25%_ncc_h2	Restructure 25%	NCC-H ₂
4	restructure_25%_ncc_elec	Restructure 25%	NCC-Electricity
5	restructure_40%_ncc_h2	Restructure 40%	NCC-H ₂
6	restructure_40%_ncc_elec	Restructure 40%	NCC-Electricity

The **Shaheen** pathway includes 6 additional S-Oil facilities from 2026, while **Restructure** pathways assume retirement of 25% or 40% of oldest NCC capacity.

4 Results

4.1 Baseline Emissions

At 70% operating rate, baseline emissions in 2025 are **47.0 MtCO₂/year**. The emission composition by fuel source:

Table 4: Baseline Emissions by Fuel Source (70% Operating Rate)

Fuel Source	Emissions (MtCO ₂)	Share (%)
Naphtha	26.58	56.5
Fuel Gas	5.96	12.7
Electricity	5.86	12.5
LNG	4.48	9.5
Byproduct Gas	2.74	5.8
Other	1.38	3.0
Total	47.00	100.0

Olefins (primarily ethylene from naphtha crackers) dominate emissions at 85.3%, making NCC decarbonization the primary focus.

4.2 Technology Deployment Logic

The model deploys technologies in cost-effectiveness order:

1. **RE-PPA** (2025+): Lowest cost; switches grid electricity to RE
2. **Heat Pump** (2025+): Replaces fossil fuel combustion (<165°C)
3. **RDH** (2026+): Addresses high-temperature BTX processes
4. **NCC-H₂ or NCC-Electricity** (2030+): Decarbonizes naphtha crackers

4.3 Critical Finding: RE-PPA and Heat Pump Synergy

A key insight is the synergistic relationship between RE-PPA and Heat Pump technologies. Heat pumps use **grid electricity** by default, creating new electricity demand. RE-PPA covers **both** baseline facility electricity **and** new heat pump electricity demand.

Table 5: Two-Step Decarbonization for Non-NCC Facilities

Step	Transformation	Emission Factor
Before	Fossil fuel combustion	~0.05-0.08 tCO ₂ /GJ
After Heat Pump	Grid electricity (COP 4.0)	0.436 tCO ₂ /MWh (2025)
After RE-PPA	Renewable electricity	0.05 tCO ₂ /MWh

This synergy enables **complete decarbonization** of non-NCC facilities by 2050.

4.4 Scenario Comparison

Table 6: Scenario Results Summary (2050)

Scenario	Emissions (Mt)	Investment (\$B)	Elec Demand (TWh)
Shaheen + NCC-H ₂	0.0	41.7	112 + H ₂
Shaheen + NCC-Elec	0.0	41.4	282
Restructure 25% + NCC-H ₂	0.0	22.8	84 + H ₂
Restructure 25% + NCC-Elec	0.0	22.5	211
Restructure 40% + NCC-H ₂	0.0	22.3	67 + H ₂
Restructure 40% + NCC-Elec	0.0	21.8	169

All six scenarios achieve Net Zero by 2050.

4.5 Regional Investment Requirements

Table 7: Regional Investment Summary (Shaheen + NCC-Electricity)

Region	Facilities	2025 Emissions (Mt)	Investment (\$B)	2050 Elec (TWh)
Yeosu	87	18.49	15.4	113
Daesan	57	15.68	13.1	95
Ulsan	85	11.94	10.2	73
Other	19	0.89	2.7	1
Total	248	47.00	41.4	282

4.6 Emission Pathway

Table 8: Emission Trajectory (Shaheen + NCC-Electricity Scenario)

Year	BAU (Mt)	Target (Mt)	Actual (Mt)	Reduction (%)
2025	47.0	47.0	47.0	0
2030	50.8	40.6	38.2	24.8
2035	53.8	35.5	30.1	44.0
2040	56.0	26.9	20.0	64.3
2045	57.4	13.4	9.1	84.2
2050	53.3	0.0	0.0	100.0

5 Discussion

5.1 Technology Selection

NCC-Electricity demonstrates **cost-effectiveness advantage** over NCC-H₂:

- Lower energy intensity: 5.0 MWh/t-ethylene vs. ~ 11.3 MWh equivalent for H₂
- No conversion losses: Direct electricity use vs. electrolysis + combustion
- Simpler infrastructure: Grid connection vs. H₂ pipeline network

However, NCC-H₂ offers strategic benefits including lower long-term energy costs (due to declining LCOH), hydrogen co-benefits for other sectors, and energy storage flexibility.

5.2 Infrastructure Requirements

The electrification pathway requires substantial grid infrastructure investment. Petrochemical electricity demand increases from ~ 8 TWh (2025) to 282 TWh (2050)—a 35-fold increase requiring approximately 40 GW of dedicated renewable capacity.

5.3 Policy Implications

1. **Grid infrastructure priority:** Massive renewable electricity expansion required
2. **Technology support:** Incentives needed for early NCC-Electricity deployment (2030 availability)
3. **Regional coordination:** Yeosu and Daesan require concentrated investment
4. **Industry restructuring:** Capacity retirement can reduce investment needs by 45%

5.4 Limitations

- Process emissions (chemical reactions) excluded—Scope 1 combustion only
- Scope 3 emissions (upstream/downstream) not analyzed
- No CCS/CCUS technologies considered
- Constant 70% operating rate assumed

6 Conclusion

This study demonstrates that **net-zero emissions by 2050 is achievable** for Korea’s petrochemical industry through:

1. NCC electrification (or hydrogen) covering $\sim 85\%$ of emissions
2. Heat pumps + RE-PPA synergy for non-NCC facilities
3. RDH technology for high-temperature processes

Investment requirements range from \$21.8B (restructuring + NCC-Electricity) to \$41.7B (Shaheen growth + NCC-H₂), with renewable electricity demand reaching 169–282 TWh/year by 2050 depending on scenario.

The critical finding is the **synergy between RE-PPA and Heat Pump** technologies: heat pumps convert fossil fuel to grid electricity, while RE-PPA decarbonizes that electricity. This two-step approach enables complete decarbonization of non-NCC facilities without requiring direct renewable electricity connection at each site.

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Data Availability

Model code and anonymized data are available at: <https://github.com/planit-institute/petrochemical-macc>

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